

Quantum-Assisted Multi-Asset Portfolio Construction

Step 06

Candidate Comparison and Explainability

Applying one audit, attribution, duplicate, and ranking framework to every portfolio

Contents

1	Why comparison needs its own methodology	3
2	Candidate registry	3
3	Common portfolio metrics	3
4	Hard constraints and soft warnings	3
5	Attribution derivations	3
6	Duplicate control	4
7	Relative component scores	4
8	Decision scenarios	4
9	Final shortlist and ranking	5
10	Mandate-dependent recommendation	5
11	Forward evidence	5
12	Pareto analysis	6
13	Deterministic narratives	6
14	Reproduction checklist	6
15	Interpretation boundary	6
A	Why comparison requires one measurement framework	6
B	Duplicate detection	6
C	Hard feasibility versus soft governance	7
D	Metric attribution	7
E	Decision-scenario ranking	7
F	Why candidate-set normalization is relative	7

G Binding guardrails	8
H Explainability narratives	8
I Final shortlist interpretation	8
J Reproduction protocol	8

1 Why comparison needs its own methodology

After several optimizers have produced portfolios, simply placing returns in one table is not enough. A fair comparison must apply the same metric definitions, constraint audit, scenario policy, attribution rules, duplicate logic, and eligibility gate to every candidate. Step 6 is that common decision layer.

2 Candidate registry

Every candidate is registered with a label, family, role, method, weight vector, source, decision eligibility, independence status, and method-traceability score. The registry prevents later tables from silently mixing diagnostic portfolios with decision candidates.

3 Common portfolio metrics

For each weight vector, Step 6 recomputes expected growth, income, total return, variance, volatility, turnover, cost, concentration, effective holdings, scenario losses, in-sample drawdown, VaR, and CVaR. Metrics are not copied blindly from each method's local output.

4 Hard constraints and soft warnings

The audit stores value, lower bound, upper bound, slack, absolute violation, normalized violation, and satisfaction. It distinguishes:

- mathematical active bounds;
- economically meaningful binding guardrails;
- trivial zero nonnegativity bounds;
- hard policy failures; and
- soft warning excess.

This distinction prevents a zero allocation from being described as a meaningful binding portfolio decision.

5 Attribution derivations

Expected return contribution is

$$RC_i = w_i(g_i + y_i), \quad \sum_i RC_i = R(w). \quad (1)$$

Let $m = \Sigma w$ be marginal variance. Variance contribution is

$$VC_i = w_i m_i, \quad \sum_i VC_i = w^\top \Sigma w. \quad (2)$$

For nonzero volatility,

$$\sigma C_i = \frac{VC_i}{\sigma(w)}, \quad \sum_i \sigma C_i = \sigma(w). \quad (3)$$

Turnover contribution is $|w_i - w_i^0|$. Linear cost contribution is $c_i |w_i - w_i^0|$. For worst scenario s^* ,

$$SC_i = w_i L_{s^*,i}, \quad \sum_i SC_i = L_{s^*}(w). \quad (4)$$

The code checks that every attribution reconciles to the portfolio total.

6 Duplicate control

Two candidates are duplicates if their final weight vectors differ by less than the declared tolerance. In the final run, QAOA, exact active-set enumeration, greedy selection, and local search all produce the same final portfolio. They receive one canonical decision representation rather than four votes. Similarly, a target-recovery audit that reproduces the primary classical portfolio is diagnostic only.

7 Relative component scores

Within the eligible set, Step 6 constructs five normalized dimensions:

1. expected-return score;
2. risk-control score;
3. implementation score;
4. guardrail-resilience score;
5. explainability score.

Risk control combines volatility, worst scenario loss, drawdown, and CVaR. Implementation combines turnover, cost, and material trades. Guardrail resilience combines policy headroom, hard-scenario headroom, warning excess, and binding count. Explainability combines method traceability, portfolio compactness, trade compactness, and attribution coverage.

8 Decision scenarios

Different mandates assign different weights to the five dimensions: Balanced, Return First, Risk First, Implementation First, Governance First, and Explainability First. For scenario r ,

$$S_{i,r} = \sum_k \theta_{r,k} s_{i,k}, \quad \sum_k \theta_{r,k} = 1. \quad (5)$$

The robust mean score averages candidate scores across declared scenarios. This is a governance aid, not an absolute performance statistic.

9 Final shortlist and ranking

Candidate	Robust mean score	Rank	Top-1 frequency	Warnings	Turnover
Strict-warning classical	0.675	1	83%	0	23.88%
Primary classical	0.559	2	17%	5	13.09%
QAOA active-rebalance	0.274	3	0%	5	11.06%

Strict warning ranks first in five of six decision scenarios. Primary classical ranks first in the return-first setting. QAOA performs most strongly in the implementation-first setting. The ranking therefore matches the economic character of the portfolios.

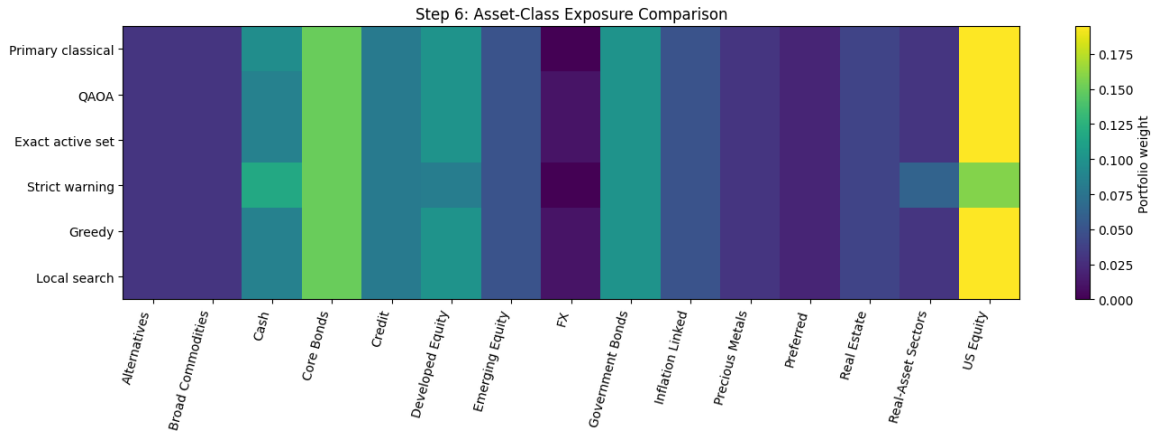


Figure 1: One of the common candidate-analysis views generated by Step 6.

10 Mandate-dependent recommendation

The correct recommendation is conditional:

- governance-first mandate: strict-warning classical;
- base-objective efficiency: primary unrestricted classical;
- hybrid-method demonstration and low-turnover active sleeve: QAOA portfolio.

No single method dominates every dimension.

11 Forward evidence

A common set of 1,000 heavy-tailed synthetic paths is attached to the three final shortlist candidates. Strict warning has the best fifth-percentile return, lowest median and tail drawdown, and lowest loss frequency. Primary classical has the highest median return. QAOA and exact active-set results are identical because their final weights are identical.

12 Pareto analysis

A portfolio is Pareto dominated if another candidate is at least as good in every selected dimension and strictly better in at least one. Step 6 reports Pareto flags for risk-return, return-implementation, governance, and comprehensive comparisons. This is more informative than forcing every decision into one scalar score.

13 Deterministic narratives

The narrative layer is generated from audited metrics: return decomposition, risk, implementation, governance status, and attribution concentration. It does not use unconstrained language generation to invent reasons. This design later supports the Step 9 co-pilot.

14 Reproduction checklist

Register all candidates before ranking, recompute common metrics, audit every hard rule, mark duplicates, define eligibility explicitly, store normalization rules and decision weights, and report rank stability. Never allow an infeasible or duplicate candidate to improve its chance through scoring.

15 Interpretation boundary

Step 6's relative scores depend on the candidate set and decision weights. They are not expected returns, probabilities, or universal investment grades. Their value is transparency: a reader can see exactly why each mandate chooses a different portfolio.

A Why comparison requires one measurement framework

Portfolios produced by different methods can only be compared fairly if every metric is recomputed from the same data, scenarios, current portfolio, and cost model. Step 6 registers candidate weights, removes duplicates, computes common metrics, applies the same hard-policy audit, and then ranks only declared eligible candidates.

The registry includes early ladder diagnostics, goal profiles, strict-warning, QAOA, exact and heuristic active-set variants, and the current baseline. Registration does not imply selection eligibility.

B Duplicate detection

Two methods may return the same weight vector. Let $w^{(a)}$ and $w^{(b)}$ be candidates. A practical duplicate rule is

$$\left\|w^{(a)} - w^{(b)}\right\|_1 \leq \epsilon_{\text{dup}}. \quad (6)$$

When QAOA, exact enumeration, greedy, and local search produce the same weights, counting all four would bias a rank aggregation. Step 6 selects one canonical representative and marks the others as duplicates. The chosen label can reflect the challenge narrative, but the numerical portfolio remains the same.

C Hard feasibility versus soft governance

A hard breach disqualifies a portfolio from a feasible shortlist. A soft warning is retained as information or a penalty. Step 6 therefore stores both

$$H_i = \text{hard breach count}, \quad W_i = \text{soft warning count}. \quad (7)$$

A portfolio with $H_i = 0$ and $W_i > 0$ is feasible under the hard policy but requires monitoring. This language is used consistently for the QAOA and primary classical portfolios.

D Metric attribution

Expected return attribution is linear:

$$R(w) = \sum_i w_i \mu_i. \quad (8)$$

Linear cost and turnover are also additive. Variance is not additive in weights alone because covariance produces pair terms. A useful Euler risk contribution is

$$RC_i = \frac{w_i (\Sigma w)_i}{\sqrt{w^\top \Sigma w}}, \quad (9)$$

which sums to portfolio volatility when volatility is differentiable and positive. The final package emphasizes total metrics and exact reconstruction; future work can extend look-through risk attribution.

E Decision-scenario ranking

There is no universally correct mix of expected return, risk, implementation, guardrail resilience, and explainability. Step 6 declares six decision scenarios, such as return-first, risk-first, implementation-first, governance-first, explainability-first, and balanced. For scenario s , score is

$$S_{i,s} = \sum_m \theta_{m,s} \widetilde{M}_{i,m}, \quad (10)$$

where $\theta_{m,s}$ are declared weights and $\widetilde{M}$ are direction-adjusted normalized metrics.

A robust summary can average ranks or scores across scenarios. The reported relative robust scores are approximately 0.675 for strict warning, 0.559 for primary classical, and 0.274 for QAOA. Strict warning ranks first in five of six scenarios; primary wins return-first; QAOA is strongest relative to peers in implementation-first.

F Why candidate-set normalization is relative

If a metric is normalized using the best and worst candidate, adding a new candidate changes every normalized score. Therefore Step 6 labels the ranking as candidate-set-dependent and preference-dependent. It is a governance aid, not a universal expected-performance ranking.

G Binding guardrails

A constraint is economically binding when its slack is close to zero and the bound represents a meaningful policy. Trivial nonnegativity bounds on zero-weight assets are excluded from the binding-guardrail narrative. This prevents a table with dozens of zero lower bounds from obscuring the constraints that genuinely shape the portfolio.

H Explainability narratives

A deterministic narrative is generated from facts in the metric and audit tables. A safe narrative follows a fixed template:

1. identify the candidate's role;
2. state hard status and warning count;
3. compare return, volatility, stress, turnover, and cost with a named baseline;
4. list the largest material trades and binding policy constraints; and
5. state limitations and evidence scope.

The text must not introduce facts that are absent from the data tables.

I Final shortlist interpretation

The three unique hard-compliant candidates represent distinct mandates:

- **Primary classical:** strongest base-objective and slightly highest expected return.
- **Strict warning:** strongest modeled downside and governance profile, with higher turnover.
- **QAOA:** sparse active-sleeve demonstration with low turnover and cost, but five soft warnings.

The shortlist does not claim that QAOA dominates classical optimization. It provides a fair method comparison and a clear reason to choose each candidate.

J Reproduction protocol

1. Register all candidates with method, role, weights, and eligibility metadata.
2. Recompute metrics from one common data context.
3. Run the same hard and soft audits on every candidate.
4. Detect duplicates before ranking.
5. Normalize metrics with declared direction and candidate set.
6. Apply every declared decision-weight scenario.
7. Export scenario ranks, robust score, guardrail rows, attribution checks, trades, and deterministic narratives.

8. Verify that changing a decision mandate can change the winner without changing any underlying portfolio.

Ranking-scope clarification

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- Step 6 scores are decision-support scores: they depend on the declared candidate set, normalization directions, and scenario weights. They are not expected returns, probabilities, or universal investment grades.
- The original Step 6 shortlist contains three candidates with common downstream evidence. Step 10 later broadens the zero-hard-breach universe to include two Step 4 portfolios; that broader screen is why the provisional Step 10 leader differs from the Step 6 shortlist leader.
- ETF-level concentration/effective-holdings measures do not capture all look-through overlap among ETFs. Economic factor or holdings overlap is a future extension.